Engine Model(s): PT6A

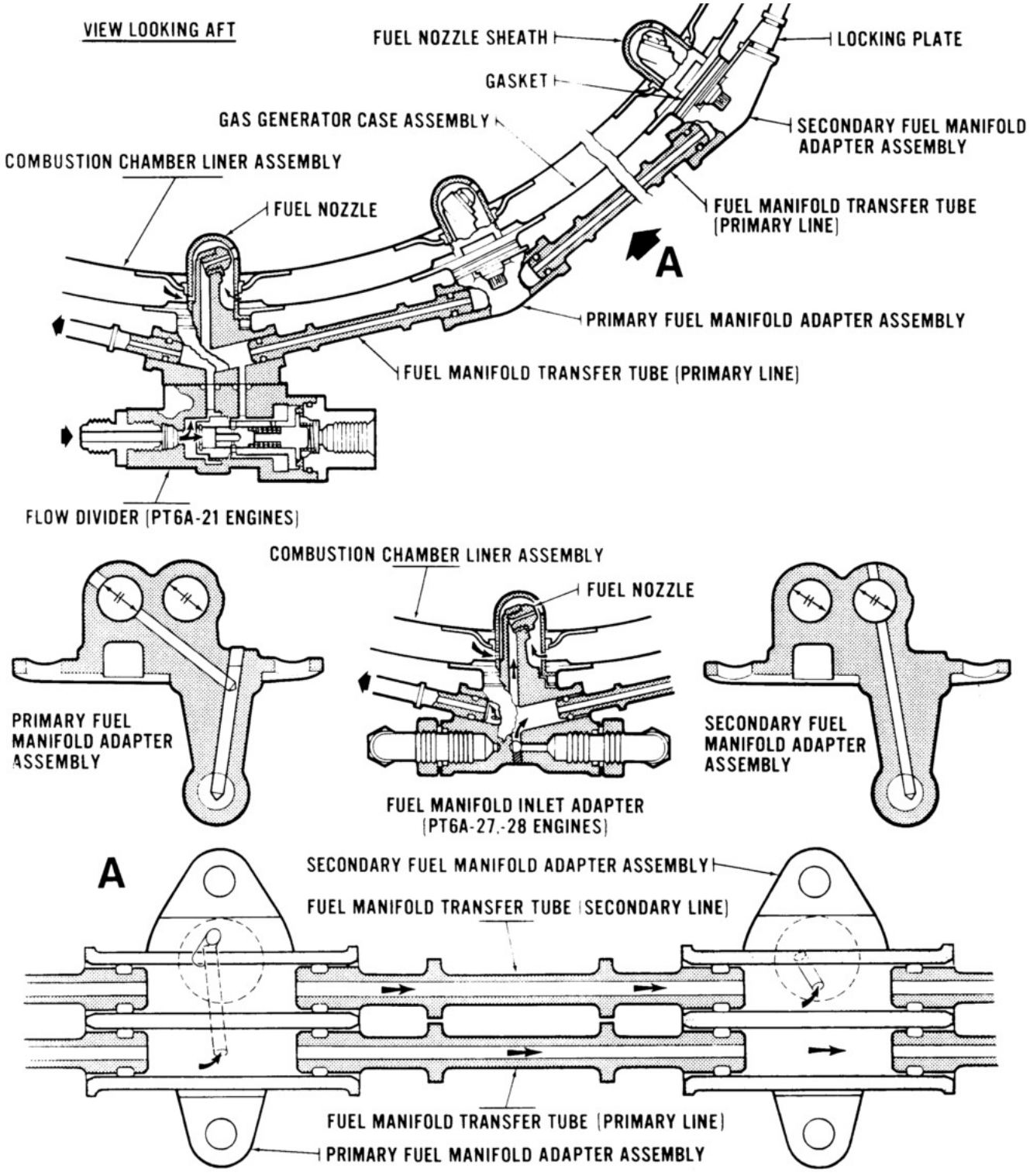
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FUEL MANIFOLD AND NOZZLES - DESCRIPTION AND OPERATION

1. Description and Operation (Ref. Fig. 1)

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Figure 1 Fuel Manifold Assembly - Cross-Section



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A. Fuel Manifold

The dual fuel manifold delivers metered fuel from the flow divider (PT6A-21) or starting flow control (PT6A-27 and PT6A-28), as applicable, to the primary and secondary fuel nozzles. The manifold consists of 14 adapter assemblies (Pre-SB1372 - seven primary, six secondary and a secondary inlet adapter; for Post-SB1372 engines - ten primary, three secondary and a secondary inlet adapter). The adapters are interconnected by pairs of fuel transfer tubes and are each secured to their respective bosses on the gas generator case by same two bolts. Locking plates, secured by the same two bolts, maintain the transfer tubes in position. The adapters and transfer tubes are sealed with gaskets and preformed packings, respectively.

B. Fuel Manifold Adapter Assemblies

Each fuel manifold adapter assembly incorporates a simplex, single orifice fuel nozzle, with swirl-type tip and a sheath. The swirl-type tips provide a fine atomized fuel spray in the annular combustion chamber liner. The sheath fits over the fuel nozzle and internal section of the manifold adapter. Each fuel nozzle assembly, which incorporates a fine strainer adjacent to the tip, is threaded into the internal section of the adapter. An internal passageway connects the nozzle to its respective primary or secondary bore in the adapter body. The sheath and nozzles extend through the gas generator case and combustion chamber liner, and are positioned so that they produce a tangential spray from one nozzle to the next in the liner. Holes in the sheath allow cooling air, from the space between the gas generator case and liner, to pass within the sheath and out through the nozzle aperture; this air, in addition to cooling the tip of the nozzle, also assists in fuel atomization.

The combustion chamber liner is located and supported within the gas generator case by the fuel nozzle sheaths. The sheaths act as spigots and pass through suspension brackets welded to the outer wall of the liner.

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